

AN EXPANSION OF MEIJER'S G-FUNCTION AND THE DISTRIBUTION OF PRODUCTS OF INDEPENDENT BETA VARIATES

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1. Introduction

In this article, a series expansion and representation in computable forms of an H-function and a Meijer's G-function, in the logarithmic case, together with the application to statistical distributions, are considered. The results obtained in this article are new results on G and H functions and the applications include the exact distributions, in the most general cases, of a good number of multivariate test statistics. The G-function discussed here is also available as the Inverse Mellin Transform of the h-th moment of a product of independent Beta variates. In section 2 we define a few Special Functions which will be used in the development of the results. Section 5 gives some verifications.

2. Some definitions and preliminary results

(i) *H-function*: The H-function is defined as,

$$(2.1) \quad H(z) = H_{p,q}^{m,n} \left[z \mid \begin{matrix} (a_1, \alpha_1), \dots, (a_p, \alpha_p) \\ (b_1, \beta_1), \dots, (b_q, \beta_q) \end{matrix} \right] = (1/2\pi i) \int h(s)z^{-s} ds,$$

where $i = (-1)^{1/2}$, z is not equal to zero and

$$(2.2) \quad z^s = \exp s (\log |z| + i \arg z)$$

in which $\log |z|$ denotes the natural logarithm of $|z|$ and $\arg z$ is not necessarily the principal value. Further,

$$(2.3) \quad h(s) = \frac{\prod_{j=1}^m \Gamma(b_j + \beta_j s) \prod_{j=1}^n \Gamma(1 - a_j - \alpha_j s)}{\prod_{j=m+1}^q \Gamma(1 - b_j - \beta_j s) \prod_{j=n+1}^p \Gamma(a_j + \alpha_j s)}$$

where p, q, n, m are integers such that,

$$(2.4) \quad 0 \leq n \leq p, 1 \leq m \leq q, \alpha_j (j=1, 2, \dots, p), \beta_j (j=1, \dots, q)$$

are positive numbers and $a_j (j=1, 2, \dots, p), b_j (j=1, \dots, q)$ are complex numbers such that,

$$(2.5) \quad \alpha_j (b_h + v) \neq \beta_h (a_j - 1 - r) \text{ for } v, r=0, 1, \dots; h=1, \dots, m; j=1, \dots, n;$$

L is a contour in the complex s -plane such that the points,

$$(2.6) \quad -s = (b_j + v)/\beta_j, \quad j=1, \dots, m; \quad v=0, 1, \dots$$

and

$$(2.7) \quad -s = (a_j - 1 - v)/a_j, \quad j=1, \dots, n; \quad v=0, 1, \dots$$

are separated and the poles in (2.6) are enclosed by L . The contour L exists on account of (2.5) and different types of contours are discussed in Braaksma (1964). An empty product is interpreted as unity. The function $H(z)$ makes sense in the following cases (2.8) and (2.9).

$$(2.8) \quad \mu \text{ is positive and } z \neq 0 \text{ where, } \mu = \sum_{j=1}^q \beta_j - \sum_{j=1}^p a_j,$$

$$(2.9) \quad \mu = 0 \text{ and } 0 < |z| < \beta^{-1} \text{ where } \beta = \prod_{j=1}^p a_j^{\alpha_j} \prod_{j=1}^q \beta_j^{-\beta_j}.$$

Fox (1961) studied $H(z)$ as a symmetrical Fourier kernel and Braaksma (1964) has given a series expansion under the condition that,

$$(2.10) \quad \beta_h(b_j + r) \neq \beta_j(b_h + v) \text{ for } r \neq h; \quad h=1, \dots, m; \quad r, v=0, 1, \dots$$

When $a_1 = a_2 = \dots = a_p = 1 = \beta_1 = \dots = \beta_q$, the H -function reduces to a G -function, see (2.11), and then the condition (2.10) is equivalent to the condition that the b_j 's, $j=1, \dots, m$ do not differ by integers. In statistical problems, especially in the problems of exact distributions of multivariate test criteria, the parameters b_j 's and a_j 's differ by integers and hence the results available in the literature are not of much help. This article also deals with a general H -function of the type $H_{p,q}^{m,0}(\cdot)$ when the conditions (2.9) are satisfied and when the parameters differ by integers.

(ii) *Meijer's G-function*: When $a_1 = \dots = a_p = 1 = \beta_1 = \dots = \beta_q$, the H -function reduces to Meijer's G -function. That is,

$$(2.11) \quad G_{p,q}^{m,n} \left[z \mid \begin{matrix} a_1, \dots, a_p \\ b_1, \dots, b_q \end{matrix} \right] = (1/2\pi i) \int_L \frac{\prod_{j=1}^m \Gamma(b_j + s) \prod_{j=1}^n \Gamma(1 - a_j - s)}{\prod_{j=1}^q \Gamma(1 - b_j - s) \prod_{j=1}^p \Gamma(a_j + s)} z^{-s} ds.$$

The existence of the contour L is given by (2.5), (2.6) and (2.7) and the existence of the function is given by the conditions (2.8) and (2.9). G -function was studied in detail by Meijer (1946) and the results were published in a series of papers. Under the condition (2.10) a series expansion is given in Erdélyi (1953, p. 208).

(iii) *The Generalized Hypergeometric Function* ${}_pF_q(\cdot)$:

$$(2.12) \quad {}_pF_q(a_1, \dots, a_p; b_1, \dots, b_q; z) = \sum_{r=0}^{\infty} \frac{(a_1)_r \dots (a_p)_r}{(b_1)_r \dots (b_q)_r} \frac{z^r}{r!}$$

where,

$$(2.13) \quad (a)_r = a(a+1) \dots (a+r-1), (a)_r = \Gamma(a+r)/\Gamma(a), (a)_0 = 1,$$

and none of the b_j 's is a negative integer or zero. The parameters are such that at least one of the expressions in (2.13) makes sense and (2.12) converges for all finite z if $p \leq q$ or $p = q+1$ and $|z| < 1$.

(iv) *The ψ -function:* The ψ -function is defined as,

$$(2.14) \quad \psi(z) = \frac{d}{dz} \log \Gamma(z) = -\gamma + (z-1) \sum_{n=0}^{\infty} \{(n+1)(z+n)\}^{-1}$$

where γ is the Euler's constant; $\gamma = 0.5772156649\dots$

(v) *The Generalized Zeta Function, $\xi(s, v)$:*

$$(2.15) \quad \xi(s, v) = \sum_{n=0}^{\infty} (v+n)^{-s}, R(s) > 1, v \neq 0, -1, -2, \dots$$

where $R(\cdot)$ means the real part of $(\cdot)$.

(vi) *The Gauss-Legendre Multiplication Formula for Gamma Functions:*

$$(2.16) \quad \Gamma(mz) = (2\pi)^{(1-m)/2} m^{mz-1/2} \prod_{r=0}^{m-1} \Gamma(z+r/m), m=2,3,\dots$$

3. A general G-function and some statistical problems

Consider a general G-function of the type,

$$(3.1) \quad G_{p,p}^{p,0} \left[z \mid \begin{matrix} a_1, \dots, a_p \\ b_1, \dots, b_p \end{matrix} \right] = (1/2\pi i) \int_L \frac{\prod_{j=1}^p \Gamma(b_j+s)}{\prod_{j=1}^p \Gamma(a_j+s)} z^{-s} ds$$

under the most general values of the parameters b_j 's and a_j 's, and examine its series expansion. The aim is to obtain a series expansion, and the representation of (3.1) in a computable form when z and the parameters are real. The conditions (2.5) for the existence of the contour and (2.9) for the existence of the function are assumed in (3.1). Further, (3.1) is equivalent to an H-function (2.1) with a_j 's and β_j 's being positive integers such that when the Gammas are expanded by using the multiplication formula (2.16) the H-function reduces to (3.1). Hence the results obtained in this article are also results on a wide class of H-functions as well. Some of the statistical problems which fall under this category will be illustrated later. In statistical problems the parameters a_j 's and b_j 's in (3.1) as well as z are real and hence for convenience it will be assumed that in what follows the parameters and z are real. However, the results obtained are valid for complex cases as well. The emphasis in this article is not on the analytical properties of (3.1) but rather it is on the series expansion and representation in computable forms. Before discussing the series expansion of (3.1) a wide class of statistical problems which

require a representation of (3.1) in computable forms, will be pointed out here.

(a₁). *Products of independent Beta variates*: If $X_1, \dots, X_p$ are independent Beta variates with the parameters (α_j, β_j) , $j=1, \dots, p$ then the h -th moment of the product $Y = X_1 \dots X_p$ is given by,

$$(3.2) \quad E(Y^h) = \prod_{j=1}^p E(X_j^h) = C_1 \prod_{j=1}^p \frac{\Gamma(\alpha_j + h)}{\Gamma(\alpha_j + \beta_j + h)}, \quad 0 < y < 1,$$

where E denotes 'mathematical expectation' and C_1 does not contain h . Since $0 < y < 1$ a moment sequence will uniquely determine the distribution of Y and hence the density of Y is given by the Inverse Mellin transform of (3.2) which is evidently of the type (3.1).

(a₂). *Ratios of Wishart determinants*: Let S_1 be a $W_p(k_1, \Sigma)$ and S_2 be a $W_p(k_2, \Sigma)$ and let them be independent. Then if $k_1 \geq p$,

$$(3.3) \quad \delta = |S_1| / |S_1 + S_2|$$

is distributed as the product of p -independent Beta variates. For the notation and proof, see, Rao (1965, p. 457). A criterion such as (3.3) arises from a general 'principle of minimum dispersion', Mathai (1967), and hence the density function of δ covers the density functions of a wide class of multivariate test criteria. Evidently the density of δ is given by a function such as the one in (3.1). Also in the following a number of likelihood ratio criteria for testing specific hypotheses in multivariate normal cases will be pointed out.

(b₁). *Wilks' criterion for testing linear hypotheses about regression coefficients*: The likelihood ratio criterion is discussed in Anderson (1958, p. 192-194) and the h -th moment of a criterion U for testing the hypotheses on regression coefficients, under the null hypothesis, is given as,

$$(3.4) \quad E(U^h) = C_2 \prod_{j=1}^p \frac{\Gamma\{(n+1-j)/2+h\}}{\Gamma\{(n+1-j)/2\}} / \prod_{j=1}^p \frac{\Gamma\{(n+1-j+q)/2+h\}}{\Gamma\{(n+1-j+q)/2\}}$$

where C_2 does not contain h . Evidently the exact distribution of U is of the type (3.1).

(b₂). *Votaw's criterion for testing compound symmetry*: Votaw (1948) has obtained the h -th moment, under the null hypothesis, of a criterion V for testing compound symmetry in the multivariate normal case, as,

$$(3.5) \quad E(V^h) = C_3 \{(p-1)^{p-1} (q-1)^{q-1}\}^h \prod_{j=0}^{p+q-3} \frac{\Gamma\{h+(n-3-j)/2\}}{\Gamma\{(p-1)[h+(n-1)/2]\} \Gamma\{(q-1)[h+(n-1)/2]\}}$$

where C_3 does not contain h and p and q are positive integers ≥ 2 . Evidently the Inverse Mellin Transform of (3.5) is an H -function. But if the multiplication formula (2.16) is used on the Gammas in the denominator of (3.5) then (3.5) can be reduced to the form,

$$(3.6) \quad E(V^h) = C_4 \prod_{j=0}^{p+q-3} \Gamma\{h+(n-3-j)/2\} / \left[\prod_{j=0}^{p-2} \Gamma\{h+(n-1)/2+j/(p-1)\} \right. \\ \left. \cdot \prod_{j=0}^{q-2} \Gamma\{h+(n-1)/2+j/(q-1)\} \right]$$

where C_4 does not contain h . Hence the exact distribution of V is evidently of the type (3.1). The exact distributions of many other test criteria such as the ones for testing sphericity, independence and so on, all fall under the category of (3.1). This is due to the fact that the test criteria or a one to one function of the criteria can be considered to be products of independent Beta variates, in a number of cases.

4. The distribution of a product of independent Beta variates

Let $X_1, X_2, \dots, X_{m+n+p}$ be a set of $m+n+p$ real independent Beta variates with the densities,

$$(4.1) \quad f_j(x_j) = \{ \Gamma(a_j + \beta_j + 1) / \Gamma(a_j + 1) \Gamma(\beta_j) \} x_j^{a_j} (1-x_j)^{\beta_j-1}, \\ 0 < x_j < 1, j=1, \dots, m+n+p, \quad a_j > -1, \beta_j > 0.$$

Then the $(s-1)$ th moment of the product $X = X_1 \dots X_{m+n+p}$ is

$$(4.2) \quad E(X_1 \dots X_{m+n+p})^{s-1} = C \prod_{j=1}^{m+n+p} \frac{\Gamma(a_j + s)}{\Gamma(a_j + 1) \Gamma(\beta_j + s)}$$

$$\text{where, } C = \prod_{j=1}^{m+n+p} \frac{\Gamma(a_j + \beta_j + 1)}{\Gamma(a_j + 1) \Gamma(\beta_j)}.$$

Now the density of X is given by the Inverse Mellin Transform of (4.2). That is,

$$(4.3) \quad f(x) = CG \left[\begin{matrix} m+n+p, & 0 \\ m+n+p, & m+n+p \end{matrix} \right] \left[\begin{matrix} a_1 + \beta_1, \dots, a_{m+n+p} + \beta_{m+n+p} \\ a_1, \dots, a_{m+n+p} \end{matrix} \right], \quad 0 < x < 1.$$

The conditions for the existence of a G-function is given in (2.9) and thus $f(x)$ exists and is uniquely determined by (4.3). The aim is to obtain $f(x)$ under all possible values of the parameters $a_1, \dots, a_{m+n+p}, \beta_1, \dots, \beta_{m+n+p}$. In general the Gamma products in (4.2) may be classified into three sets of Gamma ratios such that in the first set none of the Gamma ratios cancels out, in the second set the Gammas cancel out leaving a finite number of factors in the denominator and in the third set the Gammas cancel out leaving a few factors in the numerator. Without loss of generality, let,

$$(4.4) \quad \prod_{j=1}^{m+n+p} \{ \Gamma(a_j + s) / \Gamma(a_j + \beta_j + s) \} = \Delta_1 \Delta_2 \Delta_3$$

where

$$\Delta_1 = \prod_{j=1}^m \{ \Gamma(a_j + s) / \Gamma(a'_{1j} + \beta'_{1j} + s) \}, \Delta_2 = \prod_{k=1}^{n'} \prod_{j=1}^{q_k} (a'_{2k} + \beta'_{2k} + s - j)^{-m_{kj}}$$

and

$$\Delta_3 = \prod_{j=1}^p \{ \Gamma(a_{m+n+j} + s) / \Gamma(a'_{3j} + \beta'_{3j} + s) \}$$

where in Δ_1 none of the Gammas cancels out and in Δ_3 the Gammas cancel out leaving some factors in the numerator. In Δ_2 the general form of the factors is taken into account, where,

$$(4.5) \quad 0 \leq m_{kj} \leq n, 0 \leq n' \leq n; a'_{2i} + \beta'_{2i} - k \neq a'_{2j} + \beta'_{2j} - v$$

for all $i, j=1, 2, \dots, n', i \neq j; k, v=1, 2, \dots, \max_j \{q_j\}, m_{kj}, j=1, \dots, q_k, k=1, \dots, n'$

are all non-negative integers. For convenience, assume that none of the poles of Δ_1 coincides with any of the poles of Δ_2 or any of the zeros of Δ_3 and none of the poles of Δ_2 coincides with any of the zeros of Δ_3 . If in any particular problem, this condition is violated, it can be worked out by using the techniques developed in this article after identifying the poles properly. Hence this case will not be considered separately. Now $f(x)$ in (4.3) is available as the sum of the residues of $\Delta_1 \Delta_2 \Delta_3 x^{-s}$ at the poles of Δ_1 and Δ_2 , multiplied by C . Now consider the evaluation of these two sets of residues, with the help of the following lemmas. The proofs of these lemmas are simple and hence they are omitted.

Lemma 4.1

$$(4.6) \quad \frac{\partial^\nu}{\partial s^\nu} [\Delta x^{-s}] = x^{-s} \left\{ \frac{\partial}{\partial s} + (-\log x) \right\}^\nu [\Delta]$$

where the operator,

$$(4.7) \quad \left\{ \frac{\partial}{\partial s} + (-\log x) \right\}^\nu \equiv \sum_{j=0}^\nu \binom{\nu}{j} (-\log x)^{\nu-j} \frac{\partial^j}{\partial s^j}.$$

Lemma 4.2

$$(4.8) \quad \frac{\partial^j}{\partial s^j} [\Delta] = \left[\sum_{j_1=0}^{j-1} \binom{j-1}{j_1} D^{(j-1-j_1)} \sum_{j_2=0}^{j_1-1} \binom{j_1-1}{j_2} D^{(j_1-1-j_2)} \dots \right] \Delta,$$

where

$$(4.9) \quad D = \frac{\partial}{\partial s} \log \Delta \text{ or } \frac{\partial}{\partial s} \Delta = D \Delta \text{ and } D^{(t)} \text{ denotes the } t\text{-th partial derivative of } D \text{ with respect to } s \text{ and } D^{(0)} = D.$$

Thus from lemmas 4.1 and 4.2, it follows that,

$$(4.10) \quad G_v = \left\{ \frac{\partial}{\partial s} + (-\log x) \right\}^v = \left\{ \sum_{j=0}^v \binom{v}{j} (-\log x)^{v-j} \right. \\ \left. \sum_{j_1=0}^{j-1} \binom{j-1}{j_1} D^{(j-1-j_1)} \sum_{j_2=0}^{j_1-1} \binom{j_1-1}{j_2} D^{(j_1-1-j_2)} \dots \right\} \Delta.$$

For convenience the explicit expressions of G_v for some values of v are given below.

$$(4.11) \quad G_0 = \Delta, G_1 = (-\log x + D) \Delta, G_2 = [(-\log x)^2 + 2(-\log x) D + D^{(1)} + D^2] \Delta, \\ G_3 = [(-\log x)^3 + 3(-\log x)^2 D + 3(-\log x) (D^{(1)} + D^2) + (D^{(2)} + 3 D^{(1)} D + D^3)] \Delta, \\ G_4 = \{(-\log x)^4 + 4(-\log x)^3 D + 6(-\log x)^2 (D^{(1)} + D^2) + 4(-\log x) (D^{(2)} + 3 D^{(1)} D + D^3) + [D^{(3)} + 4 D^{(2)} D + 3(D^{(1)})^2 + 6 D^2 D^{(1)} + D^4]\} \Delta.$$

Lemma 4.3

$$(4.12) \quad \frac{\partial^{t+1}}{\partial s^{t+1}} \left[\log \frac{q}{\pi} \Gamma(a_j + s) \right] = \sum_{j=1}^q \psi(a_j + s) \quad \text{for } t = 0, \\ = (-1)^{t-1} t! \left\{ \sum_{j=1}^q \xi(t + a, a_j + s) \right\}, \quad \text{for } t \geq 1.$$

ψ and ξ functions are defined in (iv) and (v) respectively in section 2.

Now consider the residues corresponding to the poles of Δ_2 . Let R'_{kj} be the residue corresponding to the factor $(a'_{2k} + \beta'_{2k} + s - j)^{m_{kj}}$. Then,

$$(4.13) \quad R'_{kj} = \frac{1}{(m_{kj}-1)!} \frac{\partial^{m_{kj}-1}}{\partial s^{m_{kj}}} \{ (a'_{2k} + \beta'_{2k} + s - j)^{m_{kj}} \Delta_1 \Delta_2 \Delta_3 x^{-s} \}.$$

$$(4.14) \quad = \frac{x^{-s}}{(m_{kj}-1)!} \left[\sum_{u=0}^{m_{kj}-1} \binom{m_{kj}-1}{u} (-\log x)^{m_{kj}-1-u} \right. \\ \left. \times -\frac{\partial^u}{\partial s^u} \{ (a'_{2k} + \beta'_{2k} + s - j)^{m_{kj}} \Delta_1 \Delta_2 \Delta_3 \} \right], \text{ at } s = -(a'_{2k} + \beta'_{2k} - j).$$

But, at $s = -(a'_{2k} + \beta'_{2k} - j)$,

$$(4.15) \quad \frac{\partial^u}{\partial s^u} \{ (a'_{2k} + \beta'_{2k} + s - j)^{m_{kj}} \Delta_1 \Delta_2 \Delta_3 \} = \left[\sum_{v=0}^{u-1} \binom{u-1}{v} B_{kjo}^{(u-1-v)} \right.$$

$$\left. \sum_{v_1=0}^{v-1} \binom{v-1}{v_1} B_{kjo}^{(v-1-v_1)} \dots \right] W_{kjo}$$

Here (4.14) and (4.15) follow from lemmas 4.1 and 4.2 and explicit expressions of (4.14) are available from (4.11), where,

$$\begin{aligned}
 (4.16) \quad B_{kjo} &= B_{kj} \quad \text{at } s = -(a'_{2k} + \beta'_{2k} - j) \\
 &= \frac{\partial}{\partial s} \log \{ (\Delta_1 \Delta_2 \Delta_3) (a'_{2k} + \beta'_{2k} + s - j)^{m_{kj}} \} \quad \text{at } s = -(a'_{2k} + \beta'_{2k} - j) \\
 &= \sum_{i=1}^m \psi(a_i - a'_{2k} - \beta'_{2k} + j) - \sum_{i=1}^m \psi(a'_{1i} + \beta'_{1i} - a'_{2k} - \beta'_{2k} + j) \\
 &\quad - \sum_{r=1}^{n'} \sum_{\substack{i=1 \\ (r,i) \neq (k,j)}}^{q_r} [m_{ri} / (a'_{2r} + \beta'_{2r} - a'_{2k} - \beta'_{2k} + j)] \\
 &\quad + \sum_{i=1}^p \psi(a_{m+n+i} - a'_{2k} - \beta'_{2k} + j) - \sum_{i=1}^p \psi(a'_{3i} + \beta'_{3i} - a'_{2k} - \beta'_{2k} + j); \\
 (4.17) \quad B_{kjo}^{(t)} &= B_{kj}^{(t)}, \quad \text{at } s = -(a'_{2k} + \beta'_{2k} - j), \text{ for } t \geq 1 \\
 &= (-1)^{t-1} t! \left\{ \sum_{i=1}^m \xi(t+1, a_i - a'_{2k} - \beta'_{2k} + j) - \sum_{i=1}^m \xi(t+1, a'_{1i} + \beta'_{1i} \right. \\
 &\quad \left. - a'_{2k} - \beta'_{2k} + j) - \sum_{r=1}^{n'} \sum_{\substack{i=1 \\ (r,i) \neq (k,j)}}^{q_r} [m_{ri} / (a'_{2r} + \beta'_{2r} - a'_{2k} - \beta'_{2k} - i + j)]^{t+1} \right\} \\
 &\quad + \sum_{i=1}^p \xi(t+1, a_{m+n+i} - a'_{2k} - \beta'_{2k} + j) - \sum_{i=1}^p \xi(t+1, a'_{3i} + \beta'_{3i} - a'_{2k} - \beta'_{2k} + j) \}
 \end{aligned}$$

and

$$\begin{aligned}
 (4.18) \quad W_{kjo} &= W_{kj}, \quad \text{at } s = -(a'_{2k} + \beta'_{2k} - j) \\
 &= (a'_{2k} + \beta'_{2k} + s - j)^{m_{kj}} \Delta_1 \Delta_2 \Delta_3, \quad \text{at } s = -(a'_{2k} + \beta'_{2k} - j) \\
 &= \sum_{i=1}^m \pi \{ \Gamma(a_i - a'_{2k} - \beta'_{2k} + j) / \Gamma(a'_{1i} + \beta'_{1i} - a'_{2k} - \beta'_{2k} + j) \}.
 \end{aligned}$$

$$\cdot \sum_{r=1}^{n'} \frac{\pi}{\Gamma(r)} \sum_{i=1}^{q_r} \frac{\pi}{\Gamma(i)} (a'_{2r} + \beta'_{2r} - i - a'_{2k} - \beta'_{2k} + j)^{-m_{ri}}$$

(r,i) $\neq$ (k,j)

$$\cdot \sum_{i=1}^p \frac{\pi}{\Gamma(i)} \Gamma(a_{m+n+i} - a'_{2k} - \beta'_{2k} + j) / \Gamma(a'_{3i} + \beta'_{3i} - a'_{2k} - \beta'_{2k} + j)$$

Therefore, the sum of the residues corresponding to the poles of Δ_2 is,

$$(4.19) \quad \sum_{k=1}^{n'} \sum_{j=1}^{q_k} R'_{kj} = \sum_{k=1}^{n'} \sum_{j=1}^{q_k} [x^{a'_{2k} + \beta'_{2k} - j} / (m_{kj} - 1)!] \cdot$$

$$\cdot \sum_{u=0}^{m_{kj}-1} \binom{m_{kj}-1}{u} (-\log x)^{m_{kj}-1-u} \left[\sum_{v=0}^{u-1} \binom{u-1}{v} B_{kjo}^{(u-1-v)} \right.$$

$$\cdot \sum_{v_1=0}^{v-1} \binom{v-1}{v_1} B_{kjo}^{(v-1-v_1)} \dots \left. \right] W_{kjo}$$

Now we consider the poles corresponding to Δ_1 , that is, the poles corresponding to,

$$(4.20) \quad \prod_{j=1}^m \Gamma(a_j + s).$$

The Gammas in (4.20) can be classified into p_1 sets of one Gamma each, p_2 sets of 2 Gammas each, ..., p_k sets of k Gammas each, such that within each set the Gammas differ by integers for $k \geq 2$, but between the sets the Gammas do not differ by integers. This is the most general case. That is, let,

$$(4.21) \quad \prod_{j=1}^m \Gamma(a_j + s) = \prod_{r=1}^k \prod_{j=1}^{p_r} \left\{ \Gamma(a_{jr1} + s) \Gamma(a_{jr2} + s) \dots \Gamma(a_{jrr} + s) \right\},$$

for $k \geq 2$, where,

$$(4.22) \quad a_{jr1} - a_{jr2} = n_{jr1}, a_{jr2} - a_{jr3} = n_{jr2}, \dots, a_{jrr-1} - a_{jrr} = n_{jrr-1};$$

$n_{jr1}, n_{jr2}, \dots, n_{jrr-1}$ are non-negative integers and for every fixed r ,

$$a_{jri} - a_{mrh} \neq \pm v, v=0,1,\dots, j \neq m, i, h=1,2,\dots,r.$$

Now consider the Gammas,

$$(4.23) \quad \Gamma(a_{jr1}+s)\Gamma(a_{jr2}+s)\dots\Gamma(a_{jrr}+s) \quad .$$

The poles of the Gammas in (4.23) can be obtained by equating to zero, the various factors in (4.24) where the indices denote the orders of the poles.

$$(4.24) \quad \prod_{i=0}^{n_{jjr}-1} (a_{jrr}+s+i)^{\prod_{i=0}^{n_{jjr}-1} 1} (a_{jrr-1}+s+i)^2 \dots$$

$$\dots \prod_{i=0}^{\infty} (a_{jr1}+s+i)^r.$$

Let R_{vi} denote the residue corresponding to the pole of order v at $s = (a_{jrr-v+1}+i)$. Then the sum of the residues corresponding to the poles in Δ_1 is given by,

$$(4.25) \quad \sum_{r=1}^k \sum_{j=1}^{p_r} \sum_{v=1}^r \sum_{i=0}^{n_{jjr}-v-1} R_{vi}$$

with the convention that $n_{jro} = \infty$ for all $j=1, \dots, p_r$; $r=1, \dots, k$.

But,

$$(4.26) \quad R_{vi} = \lim_{s \rightarrow -(a_{jrr-v+1}+i)} \left\{ \frac{1}{(v-1)!} \frac{\partial^{v-1}}{\partial s^{v-1}} [s^{+a_{jrr-v+1}+i}]^v \Delta_1 \Delta_2 \Delta_3 x^{-s} \right\} = \frac{x^{-s}}{(v-1)!} \left\{ \sum_{h=0}^{v-1} \binom{v-1}{h} \right.$$

$$\cdot (-\log x)^{v-1-h} \frac{\partial^h}{\partial s^h} \left. \{ (s+a_{jrr-v+1}+i)^v \Delta_1 \Delta_2 \Delta_3 \} \right\},$$

$$\text{at } s = -(a_{jrr-v+1}+i) \quad (\text{by lemmas 4.1 and 4.2}).$$

Now R_{vi} can be evaluated by using the following lemmas.

Lemma 4.4. For fixed r and j ,

$$(4.27) \quad (s+a_{jrr-v+1}+i)^v \left\{ \Gamma(a_{jr1}+s) \dots \Gamma(a_{jrr}+s) \right\}, \text{ at}$$

$$s = -(a_{jrr-v+1}+i),$$

$$= \{ (n_{jr1}+n_{jr2}+\dots+n_{jrr-v-1})! (n_{jr2}+\dots+n_{jrr-v-1})! \times$$

$$\dots (n_{jrr-v-i-1})! (-1)^{v i + (v-1)n_{jrr-v+1} + \dots + n_{jrr-1}} \} /$$

$$\{ (n_{jrr-v+1} + n_{jrr-v+2} + \dots + n_{jrr-1} + i)! (n_{jrr-1-v+1} + \dots + n_{jrr-2} + i)! \}$$

$$\dots (n_{jrr-v+1} + i)! \quad i! \}$$

Proof

After absorbing the factor $(s + a_{jrr-v+1} + i)^v$ into v of the Gammas of the left hand side of (4.27), we get,

$$\begin{aligned} (4.28) \quad & (s + a_{jrr-v+1} + i)^v \Gamma(a_{jrr-1} + s) \dots \Gamma(a_{jrr} + s) \\ & = \{ \Gamma^v(a_{jrr-v+1} + s + i + 1) \Gamma(a_{jrr-1} + s) \dots \Gamma(a_{jrr-v} + s) \} / \{ (s + \\ & + a_{jrr-v+1} + i - 1)^v (s + a_{jrr-v+1} + i - 2)^v \dots (a_{jrr} + s) \}. \end{aligned}$$

This is obtained by using the simple result that,

$$(4.29) \quad \Gamma(z) = (z-1) \Gamma(z-1).$$

Now evaluating (4.28) at $s = -(a_{jrr-v+1} + i)$ and then using the result that, for a positive integer m ,

$$(4.30) \quad \Gamma(m) = (m-1)! ,$$

the lemma 4.4 is proved.

Lemma 4.5

For fixed r and j ,

$$\begin{aligned} (4.31) \quad & \frac{\partial^{t+1}}{\partial s^{t+1}} \left\{ \log \left[\prod_{n=1}^k \prod_{m=1}^{p_n} (s + a_{jrr-v+1} + i)^{p_n} \Gamma(a_{mnn} + s) \right. \right. \\ & \left. \left. \dots \Gamma(a_{mnn} + s) \right] \right\}, \text{ at } s = -(a_{jrr-v+1} + i) \end{aligned}$$

$$(4.32) \quad = \sum_{n=1}^k \sum_{m=1}^{p_n} \sum_{u=1}^n \psi(a_{mnu} - a_{jrr-v+1} - i) + v \psi(1) \\ (m, n) \neq (j, r)$$

$$+ \psi(n_{jr1} + n_{jr2} + \dots + n_{jrr-v} - i) + \psi(n_{jr2} + \dots + n_{jrr-v} - i)$$

$$\begin{aligned}
& + \dots + \psi(n_{jrr-v} - i) + v(1/1 + 1/2 + \dots + 1/i) + (v-1) \{ 1/(i+1) \\
& + \dots + 1/(i+n_{jrr-v} - i) \} + \dots + 1/(i+n_{jrr-v} + 1 + \dots + n_{jrr-1}), \\
& \text{for } t = 0 \text{ and,} \\
(4.33) \quad & = (-1)^{t-1} t! \left\{ \sum_{n=1}^k \sum_{m=1}^p n_{m=1} \sum_{u=1}^n \xi(t+i, a_{mnu} - a_{jrr-v} + 1 - i) \right. \\
& \quad + v \xi(t+1, 1) + \xi(t+1, n_{jr1} + n_{jr2} + \dots + n_{jrr-v} - i) + \xi(t+1, \\
& \quad n_{jr2} + \dots + n_{jrr-v} - i) + \dots + \xi(t+1, n_{jrr-v} - i) + v(1/(-1)^t + 1 \\
& \quad + 1/(-2)^t + 1 + \dots + 1/(-i)^t + 1) + (v-1) \{ 1/(-i-1)^t + 1 + \dots \\
& \quad + 1/(-i-n_{jrr-v} + 1)^t + 1 \} + \dots + 1/(-i-n_{jrr-v} + 1 - \dots - n_{jrr-1})^t + 1 \} \\
& \quad \text{for } t \geq 1.
\end{aligned}$$

Lemma 4.5 follows from lemmas 4.3 and 4.4. Now R_{vi} of (4.26) can be written down by using lemmas 4.4 and 4.5. That is,

$$\begin{aligned}
(4.34) \quad R_{vi} &= x_{jrr-v+1}^{a_{jrr-v+1} + i} \frac{1}{(v-1)!} \left\{ \sum_{h=0}^{v-1} \binom{v-1}{h} (-\log x)^{v-1-h} \right. \\
& \quad \left[\sum_{v_1=0}^{h-1} \binom{h-1}{v_1} A_{jro}^{(h-1-v_1)} \sum_{v_2=0}^{v_1-1} \binom{v_1-1}{v_2} A_{jro}^{(v_1-1-v_2)} \dots \right] Z_{jro} \},
\end{aligned}$$

with the convention that, $n_{jro} = \infty$ for $j=1, \dots, p$; $r=1, \dots, k$; $(v-1)!$, $\psi(v)$, $\xi(t+1, v)$ are tacitly omitted whenever v is 0, ∞ , or a negative integer, where the explicit expressions for,

$$(4.35) \quad \left[\sum_{h=0}^{v-1} \binom{v-1}{h} (-\log x)^{v-1-h} \sum_{v_1=0}^{h-1} \binom{h-1}{v_1} A_{jro}^{(h-1-v_1)} \dots \right] Z_{jro}$$

are available for some specific values of v , from (4.11) by replacing D by A_{jro} and Δ by Z_{jro} ; $A_{jro}^{(t)}$, $t \geq 1$ and Z_{jro} can be obtained by using lemmas 4.4 and 4.5. That is,

$$\begin{aligned}
(4.36) \quad Z_{jro} &= \{ (n_{jr1} + n_{jr2} + \dots + n_{jrr-v} - i - 1)! (n_{jr2} + n_{jr3} + \\
& \quad \dots + n_{jrr-v} - i - 1)! \dots (n_{jrr-v} - i - 1)! (-1)^{vi+(v-1)n_{jrr-v}+1}
\end{aligned}$$

$$\begin{aligned}
& \dots (-1)^{n_{jrr-1}}] / [(n_{jrr-v+1} + n_{jrr-v+2} + \dots + n_{jrr-1} + 1)! \\
& (n_{jrr-v+1} + \dots + n_{jrr-2} + i)! \dots (n_{jrr-1} + i)! i!] \} , \\
& \cdot \left\{ \sum_{n=1}^k \frac{\pi}{m+1} \sum_{m=1}^{p_n} \sum_{u=1}^n \frac{\pi}{u+1} \Gamma(a_{mnu} - a_{jrr-v+1} - i) \right\} , \\
& (m,n) \neq (j,r) \\
& \cdot \left\{ \sum_{k=1}^{n'} \frac{\pi}{\sigma+1} \sum_{\sigma=1}^{q_k} (a'_{2k} + \beta'_{2k} - a_{jrr-v+1} - i - \sigma) \cdot m_{k\sigma} \right\} , \\
& \cdot \left\{ \sum_{\sigma=1}^p \frac{\pi}{\sigma+1} \left[\Gamma(a_{m+n+\sigma} - a_{jrr-v+1} - i) / \Gamma(a'_{3\sigma} + \beta'_{3\sigma} - a_{jrr-v+1} - i) \right] \right\} .
\end{aligned}$$

obtained by evaluating $\Delta_1 \Delta_2 \Delta_3 (s + a_{jrr-v+1} + i)^v$, at $s = (a_{jrr-v+1} + i)$ with the help of lemma 4.4;

$$\begin{aligned}
(4.37) \quad A_{jro} &= \sum_{\sigma=1}^p \psi(a_{m+n+\sigma} - a_{jrr-v+1} - i) \sum_{\sigma=1}^p \psi(a'_{3\sigma} + \beta'_{3\sigma} \\
&\quad - a_{jrr-v+1} - i) \sum_{k=1}^{n'} \sum_{\sigma=1}^{q_k} [m_{k\sigma} / (a'_{2k} + \beta'_{2k} - \sigma - a_{jrr-v+1} - i)] \\
&\quad + \sum_{n=1}^k \sum_{m=1}^{p_n} \sum_{u=1}^n \psi(a_{mnu} - a_{jrr-v+1} - i) + v \psi(1) + \psi(n_{jr1} \\
&\quad (m,n) \neq (j,r) \\
&\quad + n_{jr2} + \dots + n_{jrr-v} - i) + \psi(n_{jr2} + \dots + n_{jrr-v} - i) + \dots + \psi(n_{jrr-v} - i) \\
&\quad + v(1/1 + 1/2 + \dots + 1/i) + (v-1) [1/(i+1) + \dots + 1/(i+n_{jrr-v+1})] \\
&\quad + \dots + 1/(i+n_{jrr-v+1} + \dots + n_{jrr-1}) ,
\end{aligned}$$

obtained as $\frac{\partial}{\partial s} \log \{ \Delta_1 \Delta_2 \Delta_3 (s + a_{jrr-v+1} + i)^v \}$, at $s = (a_{jrr-v+1} + i)$, by using lemma 4.5 and,

$$\begin{aligned}
(4.38) \quad A_{jro}^{(t)} &= (-1)^{t-1} t! \left\{ \sum_{\sigma=1}^p \xi(t+1, a_{m+n+\sigma} - a_{jrr-v+1} - i) \right. \\
&\quad \left. - \sum_{\sigma=1}^p \xi(t+1, a'_{3\sigma} + \beta'_{3\sigma} - a_{jrr-v+1} - i) - \sum_{k=1}^{n'} \sum_{\sigma=1}^{q_k} [m_{k\sigma} / \right. \\
&\quad \left. / (a'_{2k} + \beta'_{2k} - \sigma - a_{jrr-v+1} - i)^{t+1}] + \sum_{n=1}^k \sum_{m=1}^{p_n} \sum_{u=1}^n \xi(t+1, \right. \\
&\quad \left. (m,n) \neq (j,r) \right. \\
&\quad \left. a_{mnu} - a_{jrr-v+1} - i) + v \xi(t+1, 1) + \xi(t+1, n_{jr1} + n_{jr2} + \dots \right.
\end{aligned}$$

$$\begin{aligned}
& \dots + n_{jrr-v-i} + \xi(t+1, n_{jr2} + n_{jr3} + \dots + n_{jrr-v-i}) + \dots + \xi(t+1, n_{jrr-v-i}) \\
& + v(1/(-1)^{t+1} + 1/(-2)^{t+1} + \dots + 1/(-i)^{t+1}) + (v-1) [1/(-i-1)^{t+1} + \dots \\
& + 1/(-i-n_{jrr-v}+1)^{t+1}] + \dots + 1/(-i-n_{jrr-v}+1 - \dots - n_{jrr-1})^{t+1} \} \\
& \text{for } t \geq 1, \text{ which is} \\
& \frac{\partial^{t+1}}{\partial s^{t+1}} \left\{ \log [\Delta_1 \Delta_2 \Delta_3 (s+a_{jrr-v+1+i})^v] \right\}, \text{ at } s = -(a_{jrr-v+1+i}),
\end{aligned}$$

obtained by using lemma 4.5. Now by using (4.19) and (4.25) the function $f(x)$ of (4.3) can be written as,

$$\begin{aligned}
(4.39) \quad f(x) = C \left\{ \sum_{k=1}^{n'} \sum_{j=1}^{q_k} \frac{x^{a_{2k} + \beta_{2k} - j}}{(m_{kj}-1)!} \left[\sum_{u=0}^{m_{kj}-1} \binom{m_{kj}-1}{u} \right. \right. \\
(-\log x)^{m_{kj}-1-u} \sum_{v=0}^{u-1} \binom{u-1}{v} B_{kjo}^{(u-1, v)} \sum_{v_1=0}^{v-1} \binom{v-1}{v_1} \\
B_{kjo}^{(v-1, v_1)} \dots \left. \right] W_{kjo} + \sum_{r=1}^k \sum_{j=1}^{p_r} \sum_{v=1}^r \sum_{i=0}^{n_{jrr-v}} \\
\frac{x^{a_{jrr-v+1} + i}}{(v-1)!} \left[\sum_{h=0}^{n-1} \binom{v-1}{h} (-\log x)^{v-1-h} \sum_{v_1=0}^{h-1} \binom{h-1}{v_1} \right. \\
A_{jro}^{(h-1, v_1)} \sum_{v_2=0}^{v_1-1} \binom{v_1-1}{v_2} A_{jro}^{(v_1-1, v_2)} \dots \left. \right] Z_{jro} \left. \right\}, \quad 0 < x < 1,
\end{aligned}$$

where B_{kjo} , $B_{kjo}^{(t)}$, $t \geq 1$, W_{kjo} , A_{jro} , $A_{jro}^{(t)}$, $t \geq 1$, and Z_{jro} are given by (4.16), (4.17), (4.18), (4.37), (4.38), (4.36) respectively.

If $f(x)$ is written as,

$$\begin{aligned}
(4.40) \quad f(x) = C \left\{ \sum_{k=1}^{n'} \sum_{j=1}^{q_k} \frac{x^{a_{2k} + \beta_{2k} - j}}{(m_{kj}-1)!} H_{kj} \right. \\
\left. + \sum_{r=1}^k \sum_{j=1}^{p_r} \sum_{v=1}^r \sum_{i=0}^{n_{jrr-v}} \frac{x^{a_{jrr-v+1} + i}}{(v-1)!} M_v \right\}, \quad 0 < x < 1,
\end{aligned}$$

then the explicit expressions for H_{kj} and M_v can be obtained from (4.11) and further it may be observed that H_{kj} and M_v as well as $f(x)$ itself can be programmed and computed for the various values of the parameters. In writing down (4.39) the same convention as in (4.34) regarding n_{jro} , $(v-1)!$, $\psi(v)$, and $\xi(t+1, v)$, is used. The cumulative density or the distribution function,

$$(4.41) \quad F(u) = \int_0^u f(x) dx$$

is easily available by term by term integration of (4.40) with the help of the following lemma 4.6 and hence the discussion is omitted.

Lemma 4.6

$$(4.42) \quad \int_0^u x^a (-\log x)^k dx = \left[\frac{u^{a+1}}{a+1} \right] \left\{ \sum_{r=0}^k \frac{(-\log u)^{k-r}}{(a+1)^{r+1}} \right\}$$

for $a > 0, k = 0, 1, \dots, 0 < x < 1$.

This result is available by simple integration by parts and so the proof is omitted.

Thus (4.39), when x is replaced by z , ($|z| < 1$) a 's replaced by a_j 's, $a_j + \beta_j$'s replaced by b_j 's, $m+n+p$ replaced by p and omitting the C in (4.39), gives a series expansion, for the general G-function in (3.1) and the corresponding class of H-functions discussed following (3.1).

5. Applications in statistical distributions and verifications

For the purposes of illustration we will point out a few specific applications of (4.39) to exact distributions of some multivariate test statistics. Also we will verify the result (4.39) with the results available for particular parameter values.

5.1 Wilks' criterion for testing regression

The criterion and its h -th moment is given in (3.4). If the criterion is denoted as $U_{p,q,n}$ then Wilks (1932) has obtained the exact distributions for the cases $p=1,2,3, q=3$; $p=4, q=4$. Consul (1966) obtained the distribution for $p=1,2,3, q=3,4,5,6,7,8$. Pillai & Gupta (1969) have given the distribution, in multiple sums, for $p=3,4,5,6$ and have computed some percentage points which extended the results obtained by Schatzoff (1966). Mathai & Rathie (1971) have given the exact distribution in the most general cases through the techniques of partial fractions. Bartlett (1938), Box (1949) and Rao (1948) have considered some approximations.

Now (3.4) is available from (4.2). When p or q or both is even (3.4) reduces to the type (4.4) with $\Delta_1=1, \Delta_3=1$ and when p and q are both odd (3.4) reduces to (4.4) with $\Delta_3=1$. For example, when p and q are both even, (3.4) reduces to,

$$(5.1) \quad \left[\{ (s+n/2+m-2)(s+n/2+m-3)^2 \dots (s+n/2+m-1-p/2)^{p/2} \right. \\ \dots (s+n/2-1)^{p/2} \dots (s+n/2-p/2+1)^2 (s+n/2-p/2) \} \\ \left. \{ (s+n/2+m-5/2)(s+n/2+m-7/2)^2 \dots (s+n/2+m-3/2-p/2)^{p/2} \right. \\ \left. \dots (s+n/2-3/2)^{p/2} \dots (s+n/2-p/2+1/2)^2 (s+n/2-p/2-1/2) \} \right]^{-1},$$

where, $s-1 = h, q/2 = m$. That is, (5.1) is available from (4.2) by the following

substitutions: $m \sim 0, n \sim p, p \sim 0$, ($\sim$ means 'replaced by'),

$$(5.2) \quad \begin{aligned} n' &= 2, q_1 = m + p/2 - 1 = q_2, a_{21}' + \beta_{21}' = n/2 + m - 1, \\ a_{22}' + \beta_{22}' &= n/2 + m - 3/2, m_{11} = 1 = m_{21}, m_{12} = 2 = m_{22}, \dots, m_{1p/2} = p/2 \\ &= m_{2p/2}, m_{1p/2+1} = p/2 = m_{2p/2+1}, \dots, m_{1q/2} = p/2 = m_{2q/2}, \\ m_{1q/2+1} &= p/2 - 1 = m_{2q/2+1}, \dots, m_{1q_1} = 1 = m_{2q_2}. \end{aligned}$$

Under these substitutions (4.2) becomes (5.1) and hence the density of Wilks' criterion in this case is available from (4.39) by the above substitutions. For the purpose of illustration a case will be verified as well. Let $p=4, q=4$ in (3.4). In the p -even, q -even case (4.39) reduces to the form,

$$(5.3) \quad f(x) = C \left\{ \sum_{k=1}^{n'} \sum_{j=1}^{q_k} \frac{x^{a_{2k}' + \beta_{2k}' - j}}{(m_{kj} - 1)!} \left[\sum_{u=0}^{m_{kj}-1} \binom{m_{kj}-1}{u} (-\log x)^{m_{kj}-1-u} \right. \right. \\ \left. \left. \sum_{v=0}^{u-1} \binom{u-1}{v} B_{kjo}^{(u-1-v)} \sum_{v_1=0}^{v-1} \binom{v-1}{v_1} B_{kjo}^{v-1-v_1} \dots \right] W_{kjo} \right\}, 0 < x < 1,$$

where the various parameters are given in (5.2). Now when $p=4, q=4$ and under (5.2) the function in (5.3) reduces to the form,

$$(5.4) \quad \begin{aligned} f(x) &= C \left\{ x^{n/2+1-1} W_{110} + x^{n/2+1-2} [(-\log x) + B_{120}] W_{120} \right. \\ &\quad \left. + x^{n/2+1-3} W_{130} + x^{n/2+1/2-1} W_{210} + x^{n/2+1/2-2} \right. \\ &\quad \left. [(-\log x) + B_{220}] W_{220} + x^{n/2+1/2-3} W_{230} \right\}, \end{aligned}$$

where

$$(5.5) \quad C = \frac{[\Gamma(n/2+2)\Gamma(n/2+3/2)\Gamma(n/2+1)\Gamma(n/2+1/2)]}{[\Gamma(n/2) \Gamma(n/2-1/2)\Gamma(n/2-1)\Gamma(n/2-3/2)]} = \frac{[\Gamma(n+3)\Gamma(n+1)]}{[\Gamma(n-1)\Gamma(n-3)2^8]},$$

since in this case,

$$(5.6) \quad a_{21}' + \beta_{21}' = n/2 + 1, a_{22}' + \beta_{22}' = n/2 + 1/2, m_{11} = m_{21} = 1,$$

$$m_{12} = m_{22} = 2, m_{13} = m_{23} = 1, q_1 = q_2 = 3.$$

Now from (4.16) and (4.18) we obtain,

$$(5.7) \quad B_{120} = 8/3, B_{220} = -8/3, W_{110} = -8/45, W_{120} = 16/3,$$

$$W_{130} = -8/3, W_{210} = 8/3, W_{220} = 16/3, W_{230} = 8/45.$$

Under these values,

$$(5.8) \quad f(x) = x^{(n-5)/2} \left\{ \frac{\Gamma(n+3)\Gamma(n+1)}{\Gamma(n-1)\Gamma(n-3)2^8} \right\} \left\{ -8x^{5/2}/45 \right. \\ + (16/3) [(-\log x) + 8/3] x^{3/2} - 8x^{1/2}/3 + 8x^{2/3} \\ \left. + (16/3) [(-\log x) - 8/3] x + 8/45 \right\}, 0 < x < 1$$

which agrees with the results obtained by Consul (1966) and Mathai & Rathie (1969).

5.2 The sphericity test

Anderson (1958, p. 262) gives the h -th moment of a criterion W for testing the hypothesis $\Sigma = \sigma^2 I$, where Σ is the population covariance matrix, I is the identity matrix and σ^2 is some unknown quantity, in the multivariate normal case, as,

$$(5.9) \quad E(W^h) = C' p^{ph} \prod_{j=1}^p \{ \Gamma(n+1-j)/2 + h \} / \Gamma(np/2 + ph), 0 < W < 1$$

where,

$$(5.10) \quad C' = \Gamma(np/2) / \prod_{j=1}^p \Gamma((n+1-j)/2).$$

Hence the density of W is available from (5.9) as the Inverse Mellin Transform of (5.9) which is evidently an H -function, that is, the density $g(w)$ is,

$$(5.11) \quad g(w) = C' w^{-1} H_{1,p}^{p,0} \left[w/p^p \middle| \begin{matrix} (np/2, p) \\ ((n+1-j)/2, 1), j=1, \dots, p \end{matrix} \right], 0 < w < 1.$$

The exact distribution of W for $p=2$ is given in Anderson (1958) and Consul (1967) has given it for $p=3, 4$ and 6 by using some properties of Hypergeometric Functions. Now by applying the multiplication formula (2.16) on (5.9) one gets,

$$(5.12) \quad E(W^{s-1}) = K \left\{ \left[\Gamma(n/2-1-1/2+s) \Gamma(n/2-1-2/2+s) \dots \Gamma(n/2-1-p/2+1/2+s) \right] \right. \\ \left. / \left[\Gamma(n/2-1+1/p+s) \Gamma(n/2-1+2/p+s) \dots \Gamma(n/2-1+(p-1)/p+s) \right] \right\},$$

where,

$$(5.13) \quad K = \Gamma(np/2) / \{ (2\pi)^{(1-p)/2} p^{(np-1)/2} \prod_{j=1}^p \Gamma((n+1-j)/2) \}.$$

Evidently the density $g(w)$ can now be written as a G-function given in (3.1). Now when p is odd it is easy to see that the Gamma products in (5.12), excluding the constant K , can be identified with (4.4) with $\Delta_2=1=\Delta_3$. Hence the density of W is obtained from (4.39) by the following substitutions. $C \sim K$, $m \sim p-1$, $a_j = n/2 - 1 - j/2$, $j=1, \dots, p-1$; $\alpha'_{1j} + \beta'_{1j} = n/2 - 1 + j/p$, $j=1, \dots, p-1$; $p_j=0$, $j=1, 2, \dots, k-1$, $k=(p-1)/2$ and $p_k=2$. Then the density is,

$$(5.14) \quad g(w) = K \left\{ \sum_{r=1}^k \sum_{j=1}^{p_r} \sum_{v=2}^r \sum_{i=0}^{n_{jrr}-v-1} \frac{w^{a_{jrr}-v+1+i}}{(v-1)!} \right. \\ \left. \left[\sum_{h=0}^{v-1} \binom{v-1}{h} (-\log w)^{v-1-h} \sum_{v_1=0}^{h-1} \binom{h-1}{v_1} A_{jro}^{(h-1-v_1)} \sum_{v_2=0}^{v_1-1} \binom{v_1-1}{v_2} \right. \right. \\ \left. \left. A_{jro}^{(v_1-1-v_2)} \dots \right] Z_{jro} \right\}, \quad 0 < w < 1,$$

where

$r = k = (p-1)/2$, $p_k = 2$, $n_{mnu} = 1$ for all m, n and u . For example, when $p = 3$, (5.14) becomes,

$$(5.15) \quad g(w) = K w^{n/2-2} \left\{ \sum_{i=0}^{\infty} w^{1/2+i} (-1)^i \Gamma(-1/2-i) / [i! \right. \\ \Gamma(1/2+1/3-i) \Gamma(1/2+2/3-i)] + \sum_{i=0}^{\infty} w^i (-1)^i \\ \left. \Gamma(1/2-i) / [i! \Gamma(4/5-i) \Gamma(5/3-i)] \right\}, \quad 0 < w < 1.$$

In order to illustrate the simplifications of Gamma functions in similar problems we will give the simplification of (5.15) in detail. By using the formula,

$$(5.16) \quad \Gamma(z)\Gamma(1-z) = \pi/\sin(\pi z),$$

(5.15) can be written as,

$$(5.17) \quad g(w) = (K w^{n/2-2}/\pi) \left\{ w^{1/2} \sum_{i=0}^{\infty} w^i \Gamma(1/6+i) \Gamma(-1/6+i) / \right. \\ \left. [i! 2^2 \Gamma(3/2+i)] + \sum_{i=0}^{\infty} 3w^i \Gamma(-1/3+i) \Gamma(-2/3+i) / \right. \\ \left. [4 \Gamma(1/2+i) i!] \right\} = (K w^{n/2-2}/\pi) \\ \left\{ [\Gamma(1/6)\Gamma(-1/6)/4\Gamma(3/2)] F(1/6, -1/6; 3/2; w) \right. \\ \left. + [3\Gamma(-1/3)\Gamma(-2/3)/4\Gamma(1/2)] F(-1/3, -2/3; 1/2; w) \right\},$$

where, $F(\dots)$ is a Hypergeometric Function ${}_2F_1(\dots)$ defined in (2.12). Now by using the formula,

$$(5.18) \quad \Gamma(z)\Gamma(-z) = -\pi/[z \sin(\pi z)]$$

and the multiplication formula (2.16) for $\Gamma(3z)$, one obtains,

$$(5.19) \quad \Gamma(1/6)\Gamma(-1/6)/\Gamma(3/2) = -3 \pi^{1/2} 2^3$$

and

$$\Gamma(-1/3)\Gamma(-2/3)/\Gamma(1/2) = \pi^{1/2} 3^{3/2}.$$

Now by using (5.19) and then applying the result,

$$(5.20) \quad F(a, b; c; z) = (1-z)^{c-a-b} F(c-a, c-b; c; z)$$

we obtain,

$$(5.21) \quad g(w) = (K/\pi) w^{n/2-2} (1-w)^{3/2} \{ -(\pi^{1/2}/6)w^{1/2} F(4/5, 5/3; 3/2; w) \\ + (3^{5/2} \pi^{1/2}/2^2) F(5/6, 7/6; 1/2; w) \}.$$

The constant K can be simplified by using the multiplication formula (2.16). That is,

$$(5.22) \quad K = \Gamma(3n/2) \pi^{1/2} 2^{n-1} / \{ 3^{(3n-1)/2} \Gamma(n-1) \Gamma(n/2-1) \}.$$

Further, the Hypergeometric Functions in (5.21) can be combined by using the formula,

$$(5.23) \quad F(a, b; c; z) = [\Gamma(c) \Gamma(c-a-b) / \{ \Gamma(c-a) \Gamma(c-b) \}] F(a, b; a+b-c+1; 1-z) \\ + [\Gamma(c) \Gamma(a+b+c) / \{ \Gamma(a) \Gamma(b) \}] (1-z)^{c-a-b} F(c-a, c-b; c-a-b+1; 1-z),$$

$$|\arg(1-z)| < \pi,$$

for $a = 5/6$, $b = 7/6$, $c = 5/2$ and $z = 1-w$.

Now from (5.21), (5.22) and (5.23) one obtains,

$$(5.24) \quad g(w) = K_1 w^{n/2-2} (1-w)^{3/2} F(5/6, 7/6; 5/2; 1-w), \quad 0 < w < 1$$

where,

$$(5.25) \quad K_1 = 2^{n+1} \Gamma(3n/2) / 3^{(3n+1)/2} \Gamma(n-1) \Gamma(n/2-1),$$

which agrees with the result given by Consul (1967).

The above simplification is done only for the purpose of verification. For computational purposes, formula (4.39) itself can be programmed and hence further simplification may not be necessary. The exact distributions, of other multivariate test criteria which can be considered to be products of independent Beta variates, are also available from (4.39). The ψ -function in (4.39) can be simplified by using Erdélyi (1953, p. 16, (8)-(12)). In a good many cases of the parameter these ψ -functions occurring in (4.39) can be written

down, if necessary, as a finite sum. Also the ξ -function in (4.39) can be simplified by using the properties of ξ -function.

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